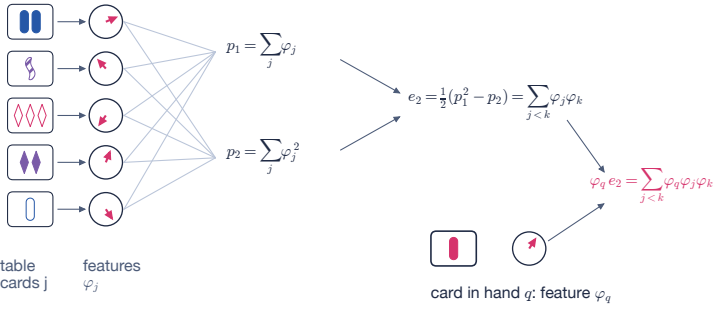
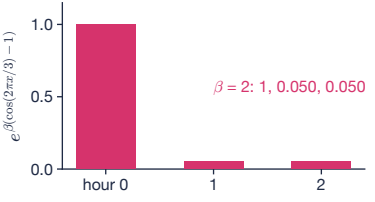
a Newton attention, one channel



trained read-out: divide by $\binom{n}{2}$ and by a root mean square over channels; a learned linear map of all 64 channels gives the answer

b Exponential Newton: one clock



$= \alpha_0 + \alpha_1 \omega^x + \alpha_{-1} \omega^{-x}$

α_0 = 0.367, $\alpha_{\pm 1}$ = 0.317

three exact terms per clock; d clocks multiply into 3^d terms, each summed over pairs by one e_2